



EQUUS
SOFTWARE

ASSIGNMENTPRO

File-Based Integration: Executing Automated Imports

This document provides an overview of how AssignmentPro™ can exchange file-based information with external systems using built-in automation capabilities combined with an Equus-hosted Secure File Transfer Protocol.

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1. Overview

AssignmentPro can automate data flow into and out of the system using file-based import and export services. For clients hosted by Equus, this capability is paired with a Secure File Transfer Protocol (SFTP) server to enable robust file-based integration solutions for connecting external systems.

The combination of file-based interfaces and SFTP is a widely used integration platform; it is trusted and well understood by corporate IT departments.

Files to be imported into AssignmentPro are generated or extracted from the source system. A non-AssignmentPro system then uploads these files to the SFTP server within Equus' secure hosting environment. Here, the **File Import Service** retrieves the files and begins the import process.

To export file-based data from AssignmentPro, the **Automated User Report Scheduler** and **File Export Service** can deposit files produced by AssignmentPro into a specified folder on the SFTP server. From there, the client or other external systems can download the files.



AssignmentPro is unable to upload or download files from a remote, client-operated SFTP server. External systems must connect to the Equus hosted SFTP server.

2. SFTP Configuration and Security Details

For hosted clients, setting up the SFTP server with an Equus-standard configuration—as described below in section “2.1 Connection Information”—is typically a straightforward process that can be completed quickly and with limited technical discussion. Advanced configuration requests may be supported but can increase the time and effort required to complete the configuration.

2.1. Connection Information

Equus can configure one SFTP server per hosted AssignmentPro instance. The SFTP hostname will use the format shown below and will mirror the URL of the associated AssignmentPro site.

sftp.[sitename].assignmentpro.com

The SFTP server port will be **9684** by default. Equus uses a non-standard port number to reduce risk and logging load. We have observed that the standard SFTP port number and common variations (e.g., 22, 2222, 8022, and so on) receive large numbers of spurious intrusion attempts. If required, the IP address and MD5 fingerprint of the SFTP server can also be provided after the server configuration has been completed.

2.2. Security Information

Equus recommends that clients request only a small number of SFTP user accounts. While user accounts can be secured with public-key authentication and traditional passwords, we recommend using public-key authentication—either alone or in combination with a password for maximum security.

An SSH2 keypair needs to be generated for each SFTP user account to establish public key authentication.

- The procedure for generating the keypair depends on the SFTP client software being used. For example, in OpenSSH, keypairs are generated using the [ssh-keygen](#) utility.
- Generate a keypair using either the ECDSA or RSA algorithms with a key size of 2048 bits or higher.
- Export only the public key in either the standard SSH2 public key format or in OpenSSH format and provide this to Equus (SSH1 or other RSA formats are unsupported.).

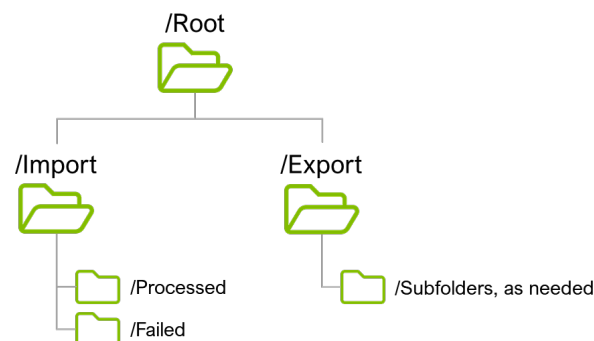
Clients will be responsible for managing key rotation schedules and submitting change requests to update the keys if required. If the client requests password authentication, Equus will create the password(s) and communicate those separately to the person responsible for maintaining the connection.

As a standard, Equus blocks Cipher Block Chaining (CBC)-based encryption, which may allow an attacker to recover the plaintext message from the encrypted data. Likewise, Equus considers MD5 and 96-bit MAC algorithms to be weak and disables them by default.

2.3. File Access Information

The standard configuration for a new SFTP server includes access to a file system structure that looks like the diagram in Exhibit 1.

Exhibit 1: A typical file system structure for a standard configuration for a new SFTP server



Data and control files to be imported into AssignmentPro should be placed in the /Import folder. These files will be moved by AssignmentPro's [File Import Service](#) to either the /Processed or /Failed subfolders as part of the import process.

Files exported from AssignmentPro are placed in the /Export folder or a corresponding subfolder. The /Export folder may have subfolders depending on client requirements and configuration. The standard SFTP file system has a total capacity of 5 GB, including import, export, and log files. Clients may specify specific file retention and deletion policy requirements if desired.

3. File Imports

AssignmentPro includes several task-specific and general-use standard imports that are available out of the box. Several “Flexible Header” imports can be tailored for an individual organization’s needs.

AssignmentPro standard imports have Equus-defined schemas and format requirements that must be followed. Generally, that means data should be formatted as a comma-separated value (CSV) file and contain a header row with column information. Documentation on individual imports can be provided on request.

Import files can be uploaded manually through the AssignmentPro application’s [Import Data](#) screens, or the [File Import Service](#) component can be used to automate this process. This service periodically scans a configured /Import folder looking for import control files to process.

Import control files are unencrypted XML-based files. These files contain metadata that defines the settings necessary for the import processor to handle and import a data file correctly.

When the [File Import Service](#) scans a folder and finds valid control files, it will read these files in oldest-first order. The service will then use the XML control files' metadata to queue the corresponding data files to be imported into AssignmentPro.

3.1. Import Content and Timing

During initial AssignmentPro implementation, imports may be used to bring large amounts of data into AssignmentPro. After this period, we recommend limiting the importation of unchanged and duplicate data.

There may be performance and hardware resource limitations for large import files. Where possible, we recommend that import files contain data for changed records only—sometimes called a “delta” file.

The frequency that data should be uploaded for import depends on the business process being supported. For example:

- New assignee initiations could occur as needed in a near-real-time data flow.
- Employee demographic updates may occur in a nightly batch.
- Payroll-related information may occur biweekly.

Each import uses server resources. Processing, especially large or frequent imports, may affect the site's performance. Therefore, import schedules should be designed to balance performance issues and the need for timely data.

3.2. Creating a File Import

An automated import is composed of two required files:

1. An **import data file** (i.e., .CSV, .DAT, .PGP, and similar formats) that composes the data to import. The file extension of the data file **cannot** be .XML.
2. An **import control file** with an .XML extension that holds the settings necessary to process the import



IMPORTANT NOTE: Both files will need to be uploaded to the import folder, but **the data file always needs to be uploaded first**. There is a setting in the control file that specifies the data file's name. If the specified data file is not present when the XML control file is read, the import will fail.

The name of the import data file can be any valid Windows file-system filename up to 255 characters. Except it cannot contain a .XML extension, which is reserved for the import control file. The file's upload date, not the name, is used to determine the order in which files will be processed, with the oldest file being processed first (i.e., First In, First Out or FIFO.)

Equus generally recommends that the data file names contain a date and time stamp. This time stamp will protect against overwriting import files when previously uploaded files haven't been processed for import yet. This approach is merely a suggestion; some clients never vary the import file names.

Whatever name is given to the import data file, that name needs to be reflected in the **"fileName"** attribute of the accompanying XML control file. Likewise, we recommend the control file follow the same naming scheme as the data file, just with a .XML file extension.

For example, to import a file of employee demographic data updates, the two files uploaded might be named something like this:

Import Data File ▶	Employee_Update_201508011200.CSV.PGP
Import Control File ▶	Employee_Update_201508011200.XML

3.3. PGP File Encryption

AssignmentPro supports decrypting PGP-encrypted files that are uploaded for import. Equus recommends that PGP encryption be applied to any data file containing personally identifiable information (PII), such as employee demographic imports or compensation information. Generally, data files containing other information, such as the exchange rate or company organization update files, do not need to be encrypted.

If PGP encryption on import files is requested, an Equus technician will generate a 2048-bit PGP keypair and provide the public key to be used by the encrypting application. Clients may specify if a higher bit length is desired. Clients may also request other PGP keypair parameters, but these may not be compatible with AssignmentPro.

3.4. Import Control File

The import control file is required since it contains the information necessary for AssignmentPro to process an import. It is an XML file and must conform to a prescribed structure communicated by Equus.

Exhibit 2 is an example of an import control file accompanying an Employee/Assignment Flexible Header Import data file. This import has no parameters but uses encryption. Exhibit 3 is an example control file for a Compensation Accumulator import without encryption.

Exhibit 2: Sample import control file for an Employee/Assignment Flexible Header import using encryption

```
<?xml version="1.0" encoding="utf-8" ?>
<Import
  id="19"
  company="1"
  userName="HRIS File Import"
  externalID="E10001"
  description="Daily employee demographic update file"
  fileName="Employee_Update_20190831.csv.pgp"
  errorToEmailAddress="optional_error_monitor_mailbox@mycompany.com"
  cryptoConfigName="Equus_Provided_Encryption_Key_Name" />
```

Exhibit 3: Sample control file for a Compensation Accumulator import

```
<?xml version="1.0" encoding="utf-8" ?>
<Import
  id="22"
  company="1"
  userName="Admin"
  externalID="E10001"
  description="Expense file for compensation accumulation"
  fileName="ExpCompAccumulator_20190831.csv"
  errorToEmailAddress="optional_error_monitor_mailbox@mycompany.com">
  <Parameters>
```

```
<Parameter name="@DATA_SOURCE" value="Expense Export Process"></Parameter>
<Parameter name="@EXCHANGE_RATE_DATE_PREF" value="EQCAFL"></Parameter>
<Parameter name="@EXCHANGE_RATE_DATE" value="1900-01-01"></Parameter>
</Parameters>

</Import>
```

The first line of a control file can be a standard declaration that it is an XML document and include the version of XML being used, as is depicted in the previous two exhibits. The XML declaration can be omitted if necessary.

The root element of the control file must be a single **<Import>** node. The **<Import>** node will contain attributes that direct how the **File Import Service** and the **AssignmentPro Import Engine** will handle the file import. These attributes are listed and described in the following table.

FIELD	REQUIRED OR OPTIONAL	DESCRIPTION
ID	Required	This integer identifies the type of import to be performed. It must correspond with the ID of a particular import from the Import Settings screen in AssignmentPro. If the value is not numeric or missing, the import will fail. An error will be logged in the Windows Event Log. Equus will provide information on the correct value for a particular import.
company	Required for some imports	This integer value should correspond to the ID field in the COMPANY table. It should be set if it is a company-level import. Omitting this attribute or leaving the value empty will indicate to the import engine that this is a system-level import. This field must be numeric. If it is not numeric, the import will fail and an error will be logged in the Windows Event Log. AssignmentPro supports both system-level and company-level imports. Equus will provide information on the correct value for a particular import.

FIELD	REQUIRED OR OPTIONAL	DESCRIPTION
externalID	Optional	The attribute's value is written to the Import History External ID field. It can be included if you want to tie an export job ID from your system to the import job history record in AssignmentPro.
description	Optional	The attribute's value is written to Import History Description [???]. This is recommended when the same type of import files are uploaded from different systems or with different conditions.
userName	Required	The username associated with the import; it will be displayed on the import history screen.
fileName	Required	This field is the name of the data file that contains the records to be imported. If this file does not exist, the import will fail, and an error will be logged on the hosting server where an Equus technician will be able to retrieve it and troubleshoot the problem.
errorToEmailAddress	Optional	An email can be sent if the import fails. This is the email address to which the error email will be sent. You can provide multiple email addresses here by separating them with a comma.
cryptoConfigName	Optional	If PGP encryption is employed, this attribute must contain a reference to the cryptographic key to decrypt the file when the import is processed. Equus will provide information on the correct value for a particular import.

FIELD	REQUIRED OR OPTIONAL	DESCRIPTION
writeLogFile	Optional	This attribute should be excluded unless per-import log files are required.

Some imports require parameters to be passed to the **AssignmentPro Import Engine** so the data can be imported correctly. In these cases, one **<Parameters>** node is required, and within the **<Parameters>** node there will be as many child **<Parameter>** nodes as needed for the particular import. Equus will help determine the parameter requirements for individual imports.

For imports that don't require parameters, the **<Parameters>** node and its children should be omitted and the **<Import>** node closed. This import will be illustrated as a single tag with a trailing slash like "**<Import id=1 ... />**."



IMPORTANT NOTE: A **<Parameter>** node must exist and contain a value for every parameter associated with an import, even if it is optional or conditionally displayed on the **Import Data** screen in AssignmentPro. For example, in Exhibit 2, the @EXCHANGE_RATE_DATE parameter contains a placeholder value of "1900-01-01."

3.5. Import File Processing

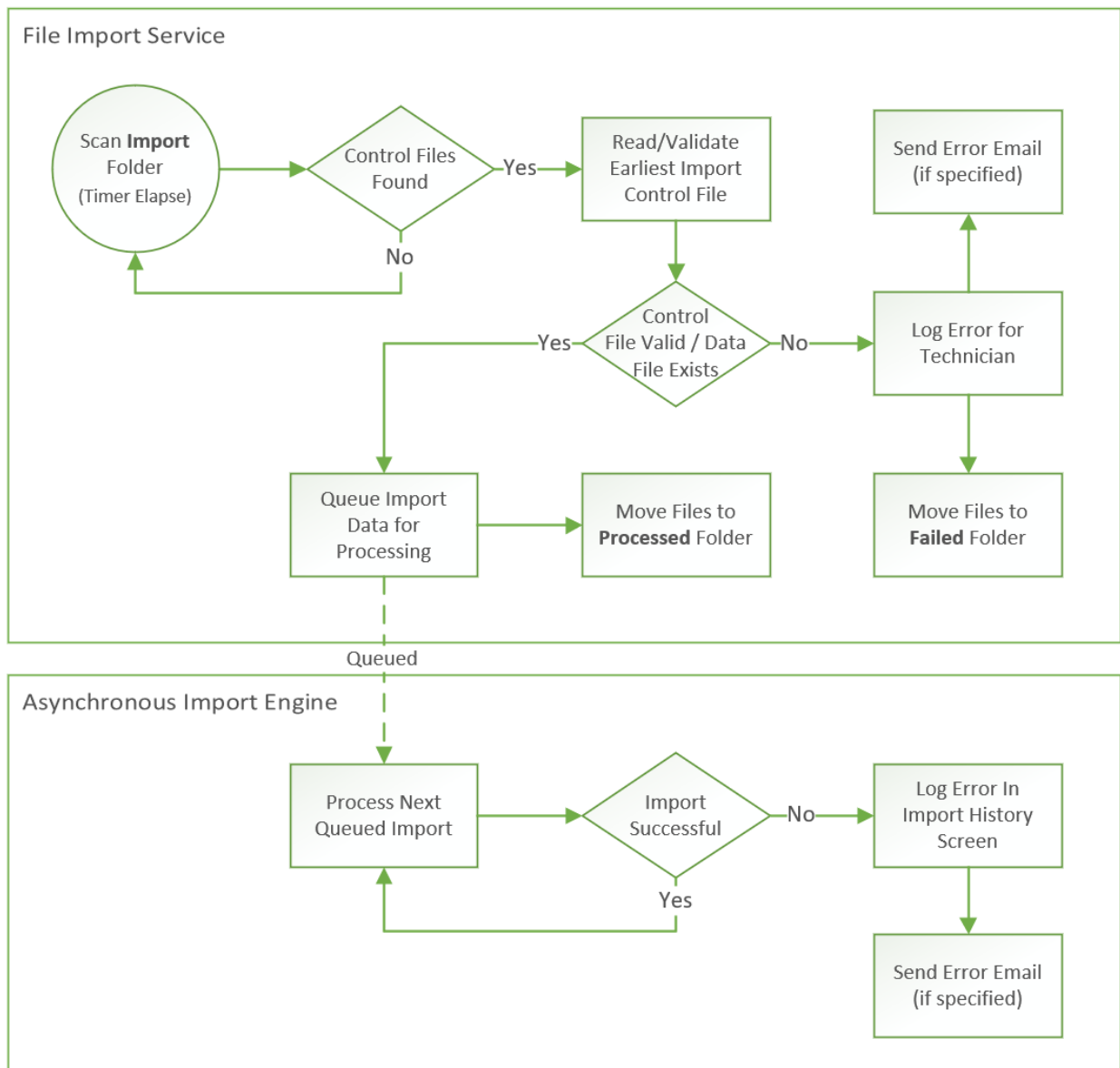
The **File Import Service** (illustrated in Exhibit 4) will periodically scan the /Import folder and look for new import control files with an .XML file extension. The service will process new control files in order from oldest to most recent upload date. If the control file is successfully read and the corresponding data file is present in the /Import folder, the import data will be queued for import processing.

At that point, the import files will be moved to the /Processed folder, which is typically a subfolder of the /Import folder. If the **File Import Service** fails to queue the import data for further processing, the import control file and data file will be moved to the /Failed folder, typically a subfolder of the /Import folder.

The import can be successfully "processed" by the **File Import Service** and still fail to be imported by AssignmentPro. The actual data import process is handled asynchronously by a different service.

When uploading the two files necessary for an import, the data file should be uploaded first and the control file second. This approach avoids the situation where a control file references a data file that does not yet exist on the server.

Exhibit 4: Flow diagram of the file import process



4. File Exports

There are three mechanisms by which data can be extracted from AssignmentPro automatically and placed as files on the SFTP server for download.

An Equus technician must activate the appropriate export services and configure the SFTP service before any of these methods can be employed.

4.1. User Reported-Based Exports

The most common method is to create a **User Report** in AssignmentPro; configure its export settings; and set the report to be exported to a file on a scheduled, recurring basis, daily, weekly or monthly. User Report-based exports can:

- Be configured to export in CSV format, with a selectable separator character.
- Have customizable header and footer rows.
- Be run with the same data security level as the user who sets up the export job.

4.2. Payroll-Related Data Exports

Exports of payroll-related data associated with AssignmentPro's **Compensation** module and the **Compensation Control Panel** can be initiated after the **Payroll System Reporting Batch** process.

The exported files and their names, formats, and encryption key can be specified distinctly for each configured payroll system on the **Payroll System – Reporting** tab in AssignmentPro.

Please review the AssignmentPro **Compensation** module documentation or contact your account team for more information on configuring payroll system report batches.

4.3. Standard Exports

Finally, AssignmentPro has a limited number of standard exports that can be accessed through the **Export Data** screen.

Most clients who use these exports do so on an ad-hoc basis through the AssignmentPro user interface. It is rare for a client to request these exports to be automatically generated and placed on the SFTP server.

5. PGP File Encryption

AssignmentPro can apply PGP encryption to files exported using the standard export methods.

Equus recommends that encryption be applied to any data file containing PII. To enable file encryption, Equus will apply a client-provided PGP public key to AssignmentPro so it can be used to encrypt export files. If the encryption key has an expiration date, the key provider should track the date to prevent interruptions caused by expired keys.

Clients are responsible for managing key rotation schedules and submitting change requests to update PGP keys if necessary.

SUMMARY

Rich data, combined with innovative technology tools, helps connect the entire mobility value chain of HR, Talent, Reward, and vendor systems to facilitate the seamless management of a company's global workforce. Built by a team with a passion for global mobility, Equus solutions provide integrated, end-to-end functionality, performance, and ease-of-use that is unmatched in the industry.

Harness the power of connectivity to simplify the mobility process, gain strategic insights, and enhance the employee experience with the Equus AssignmentPro. Global Mobility. Connected.

ABOUT EQUUS

Equus Software is the global leader in cloud-based international relocation and mobility solutions. More than 300 organizations around the world rely on Equus tools and technology to automate mundane, transactional work so that global mobility teams, talent management professionals and other key stakeholders can focus on adding value to the business. Founded in 1999, Equus has a proven track record for delivering cutting-edge talent mobility solutions, continuous innovation, and exceptional customer service.

 **EQUUS**
SOFTWARE
Inquiries@equusoft.com
+1 (303) 292-4200
www.equusoft.com